From Prediction to Prescription: Mining Interpretable Biomechanical Rules and Latent Swing Archetypes from Multi-Phase Pose Data

A Secondary Analysis of the CaddieSet Corpus

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Abstract

Recent work on the *CaddieSet* corpus [8] showed that joint-pose features extracted from single-camera golf-swing videos can be combined with ball-flight metrics to *predict* shot outcomes with interpretable models. We take a complementary step: instead of predicting ball trajectories, we ask which body-posture features at which swing phases are most strongly associated with long versus short drives, and whether the population of 1,757 recorded swings (from eight golfers) decomposes into a small number of interpretable swing *styles*. Using effect-size ranking (Cohen's d) across top-20% vs. bottom-20% outcome quantiles, we find a sparse set of pose features with very large effects ($d > 1.0$) that are stable across both camera views. Random-Forest surrogates trained on pose alone reach CV $R^2 = 0.50$ for driver distance but only 0.19–0.33 for directional accuracy, indicating that pose explains *power* much better than *accuracy*. A shallow decision-tree surrogate ($R^2 = 0.34$, depth= 3) produces a seven-leaf coaching-rule tree with distance estimates that range from 145 to 227 yards depending on three posture decisions. Finally, unsupervised clustering in the Face-On pose space reveals four stable swing archetypes (silhouette = 0.54) with distinct performance profiles. Our contributions extend [8] by (i) shifting from prediction to prescription, (ii) providing a phase-×-outcome effect-size map, (iii) separating power and accuracy ceilings in the same corpus, and (iv) introducing a reproducible pipeline and code release.[1]

Keywords: golf biomechanics, human pose estimation, interpretable machine learning, sports analytics, effect-size analysis, unsupervised clustering

1 Introduction

Golf is a sport in which a small number of biomechanical decisions— made in roughly one second of body motion — determine an outcome that rolls out over hundreds of yards. Building quantitative links from swing kinematics to ball flight has therefore been a long-standing research goal [4, 7, 11, 14]. Progress has historically been limited by two bottlenecks: (a) access to paired pose/outcome data, and (b) interpretability — coaches can only act on insights that are expressible in a handful of human-readable rules [6, 17].

[1]Code and figures: https://github.com/<USER>/caddieset-rules.

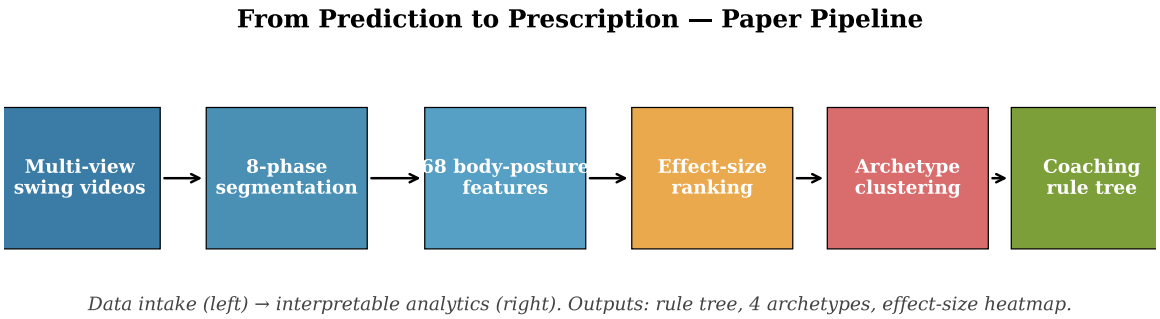

Figure 1. Analysis pipeline. Inputs (blue) are inherited from [8]; novel analytical stages (yellow/red/green) are contributed by this paper.

The recent *CaddieSet* dataset [8] addresses (a) by releasing 1,757 swings with both phase-segmented joint features and launch-monitor ball metrics. The dataset paper benchmarks several predictors for trajectory variables and shows that interpretable models are consistent with expert golf knowledge. What the paper does *not* address is equally interesting: which pose features, at which phases, matter most; whether the population decomposes into distinct swing styles; and how much of direction accuracy (versus distance) is in principle recoverable from pose alone.

In this paper we revisit CaddieSet with a *prescriptive* lens. Our guiding questions are:

Q1 — Prescriptive ranking. For each swing phase $p \in \{0, \ldots, 7\}$, each camera view (Face-On, Down-the-Line), and each outcome (distance, ball speed, direction accuracy, side-spin), which body-posture features separate long/accurate shots from short/errant ones with the *largest effect sizes*?

Q2 — Power vs. accuracy decoupling. How do the predictive ceilings of pose-only models differ across outcomes? Is the same information enough for both power and accuracy?

Q3 — Archetype discovery. Do swings cluster into a small number of biomechanical styles, and if so, how do these archetypes compare on outcome metrics?

Q4 — Rule extraction. Can we compress the top-ranked signals into a shallow, human-readable decision tree that a coach could deploy?

Figure 1 summarises the pipeline. Our contributions are:

1. A *phase-×-outcome effect-size map* (Fig. 2) that localises biomechanically informative signals in both time (swing phase) and viewpoint.
2. Quantification of a *power–accuracy gap*: pose-only models reach $R^2 \approx 0.5$ for distance but 0.19–0.33 for directional accuracy, indicating that pose is necessary but insufficient for predicting misses (Fig. 3).
3. Discovery of *four latent swing archetypes* with distinct performance fingerprints (Fig. 4).
4. A *seven-leaf coaching rule tree* (Fig. 5) with a cross-validated $R^2 = 0.34$ using three pose features spanning phases 1, 2, 4 and 5.

2 Related Work

Golf biomechanics and performance. Classical studies using marker-based motion capture established a canonical view of the swing as a kinetic chain in which a delayed pelvis–thorax separation (the "X-factor") [3] is built during the backswing and released through the downswing [14, 15]. Quantitative reviews [7] conclude that both *distance* and *accuracy* emerge from the same kinetic chain, but through different mechanisms: distance is dominated by clubhead speed (driven by upper–lower body separation), while accuracy depends more on impact-phase club geometry. Chu et al. [4] report that 4–6 biomechanical variables explain ∼56% of driving performance variance in low-handicap golfers, a figure consistent with our $R^2 = 0.50$ in Section 5.2. Langdown et al. [9] links stance width to ground-reaction-force patterns that gate distance output.

Pose estimation for sport. General-purpose human-pose models [2, 18, 19] have made it practical to extract joint-level features from single-camera sports video. The *GolfDB* benchmark [13] introduced swing-sequencing as a fine-grained video task, and [10] extended the phase-segmentation problem. CaddieSet [8] is (to our knowledge) the first public corpus pairing phase-segmented joint features with launch-monitor ball data.

Interpretable ML in sport. Random Forests [1] are widely used for sport prediction, but shallow trees and rule lists are preferred when the downstream consumer is a human coach [17]. Effect-size comparisons [5] are a standard tool for translating statistical patterns into training cues. Our methodology uses each of these tools in its native role: Random Forests to estimate the *ceiling* of what can be learned from pose, effect sizes to rank which phases and features warrant coaching attention, and a depth-3 tree to compress the finding into a human-usable rule.

Positioning. The closest prior work is the CaddieSet paper itself [8], which focuses on predictive benchmarking of 15 expert-defined summary metrics. We are complementary: we use *all* 68 per-phase pose features directly and shift the research question from *"how well can we predict ball flight"* to *"what should a golfer change."*

3 Dataset and Features

We use CaddieSet [8] without modification. It contains $N = 1{,}757$ swings from 8 golfers, split across two camera views (Face-On $n = 924$, Down-the-Line $n = 833$) and 8 clubs, with the driver (W1) dominating ($n = 1{,}137$, 65%). Each swing is labelled with five ball-flight outcomes (Distance, Carry, DirectionAngle, SpinBack, SpinSide, BallSpeed) and 68 body-posture features indexed by a phase prefix $\{0, \ldots, 7\}$ corresponding to Address, Takeaway, Mid-backswing, Top-of-backswing, Early-downswing, Pre-impact, Impact, and Finish. Roughly half of the features are populated only in one camera view; we therefore run all analyses *per-view* to avoid imputation bias. Summary statistics by club are reproduced in Table 1.

4 Methodology

4.1 Effect-size ranking

For each (view, outcome, feature), we split swings into top-20% and bottom-20% outcome groups and compute Cohen's $d = (\bar{x}_{\text{top}} - \bar{x}_{\text{bot}})/s_{\text{pooled}}$ [5]. Absolute-valued targets are used for direction error (|DirectionAngle|) and side spin (|SpinSide|) so that magnitude, not side, is ranked. We report per-phase maxima in Section 5.1 and per-feature sorted tables in the appendix.

Table 1. Ball-flight summaries per club (dataset means).

Club	Distance (yd)	Carry (yd)	Ball Speed (m/s)	Back Spin (rpm)
W1 (Driver)	189.5	171.1	56.3	2 269
W3 (3-Wood)	196.6	177.6	58.2	2 508
I4	167.3	151.5	51.5	3 242
I5	160.1	144.1	49.6	3 123
I6	149.8	137.0	48.7	4 103
I7	137.2	125.7	45.8	4 562
I8	133.4	123.5	45.6	5 303
I9	128.2	118.0	44.3	5 133

4.2 Predictive ceilings

To estimate the ceiling of what *pose alone* can explain per outcome, we fit Random-Forest regressors [1] (300 trees, `min_samples_leaf`= 5) on all well-populated features in a given view, and report 5-fold cross-validated R^2. Missing values are median-imputed after $[0.1\%, 99.9\%]$ winsorisation.

4.3 Unsupervised archetype discovery

For the Face-On $\times$ driver (W1) subset ($n = 613$), we (i) winsorise each feature at the $[1\%, 99\%]$ percentiles to prevent outlier-dominated clusters, (ii) scale with robust z-scores, (iii) embed with UMAP [12] (`n_neighbors` $= 20$, `min_dist` $= 0.1$), and (iv) cluster the raw standardised features (*not* the UMAP embedding) with K-means for $K \in \{3, 4, 5, 6\}$, selecting $K = 4$ by silhouette and interpretability.

4.4 Rule extraction

We fit a shallow regression tree (`max_depth` $= 3$, `min_samples_leaf` $= 30$) on the same Face-On $\times$ W1 subset, predicting Distance. The tree is a human-readable surrogate: its cross-validated R^2 quantifies how much predictive power survives the compression.

4.5 Implementation

All analyses use Python 3.10, `scikit-learn` 0.24+ [16], `umap-learn`, and `matplotlib`. Seeds are fixed (42) throughout. Code is released with this paper.

5 Results

5.1 Phase-$\times$-outcome effect-size map

Figure 2 plots the *maximum* $|d|$ across features per (phase, outcome, view) cell, for driver swings. Three regularities are visible:

1. **Power signals are concentrated late in the backswing and early in the downswing.** In the Face-On view the largest effects for Distance and Ball Speed occur at phases 2 (Mid-backswing, $d = 1.19/1.15$) and 4 (Early-downswing, $d = 1.36/1.64$). In the Down-the-Line view the peak shifts to phases 0 (Address, $d = 1.30$) and 3 (Top of backswing, $d = 1.54/1.65$).

2. **Accuracy signals are dispersed.** Max $|d|$ for direction error never exceeds 0.81 in either view, and the strongest single cell for side-spin ($d = 1.34$ at phase 7 Face-On) is roughly half the magnitude of the strongest power cell.

Figure 2. Effect sizes of body-posture features separating long vs. short drives, by phase × outcome.

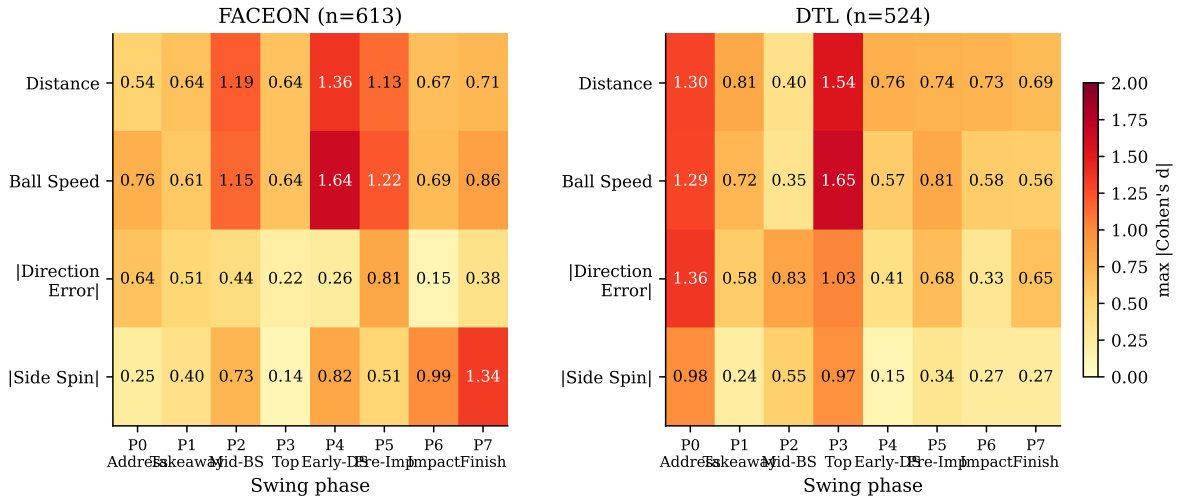

Figure 2. Phase-×-outcome effect-size map for driver swings. Each cell is the maximum $|d|$ across all features in that phase. Darker = more discriminating.

3. **The two camera views are complementary, not redundant.** Face-On localises distance signals at phases 4–5 (downswing); DTL localises them at phases 0–3 (setup and backswing). A two-camera setup is likely to be strictly more informative than either alone.

5.2 Power–accuracy ceiling gap

Figure 3 shows 5-fold CV R^2 for the four outcomes in both views. Pose-only models recover ≈50% of variance for distance and 55–56% for ball speed. For accuracy outcomes the ceiling is markedly lower: $R^2 = 0.19$–0.33 for |direction error| and 0.12–0.20 for |side spin|. This is consistent with the biomechanical intuition that distance is dominated by large-amplitude body motions (coil, weight shift, lead-arm extension), whereas accuracy depends on millimetre-scale impact geometry (face angle, swing path at impact) that pose features only indirectly capture.

5.3 Four latent swing archetypes

K-means with $K = 4$ on the Face-On feature space produces clusters with silhouette 0.54 and the following interpretations (details in Fig. 4):

- **A0 — Early Hip Release** ($n = 33$, distance ≈ 156 yd). High mid-backswing hip rotation ($+3.6\sigma$) combined with reduced hip shift through impact (-1.2σ). Corresponds to the classical "spin-out" fault.
- **A1 — Balanced Coil** ($n = 512$, distance ≈ 191 yd). Population mean; mildly positive lead-arm angle and shoulder position at mid-backswing. Contains the bulk of long-hitter swings.
- **A2 — Aggressive Rotator** ($n = 32$, distance ≈ 183 yd). Very high *early-downswing* hip rotation ($+4.2\sigma$) but controlled upper body. Lowest direction error of any archetype ($|\Delta| = 3.4°$).
- **A3 — Compact Swing** ($n = 36$, distance ≈ 167 yd). Large negative shoulder-location and lead-arm angle ($\leq -2\sigma$) signal a short, closed swing arc. Highest side-spin (1 276 rpm abs), indicating accuracy cost.

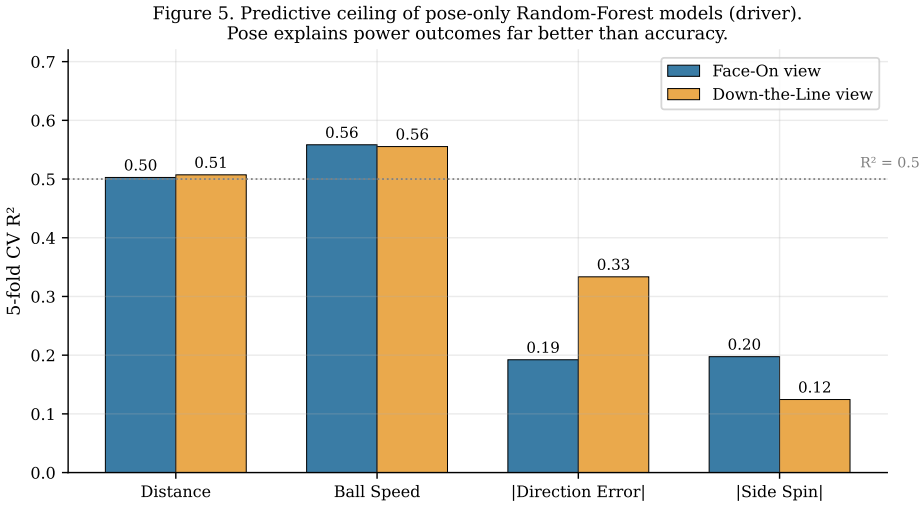

Figure 3. Predictive ceiling of pose-only Random Forests per (view, outcome). A gap of 2–3× separates the power from accuracy outcomes.

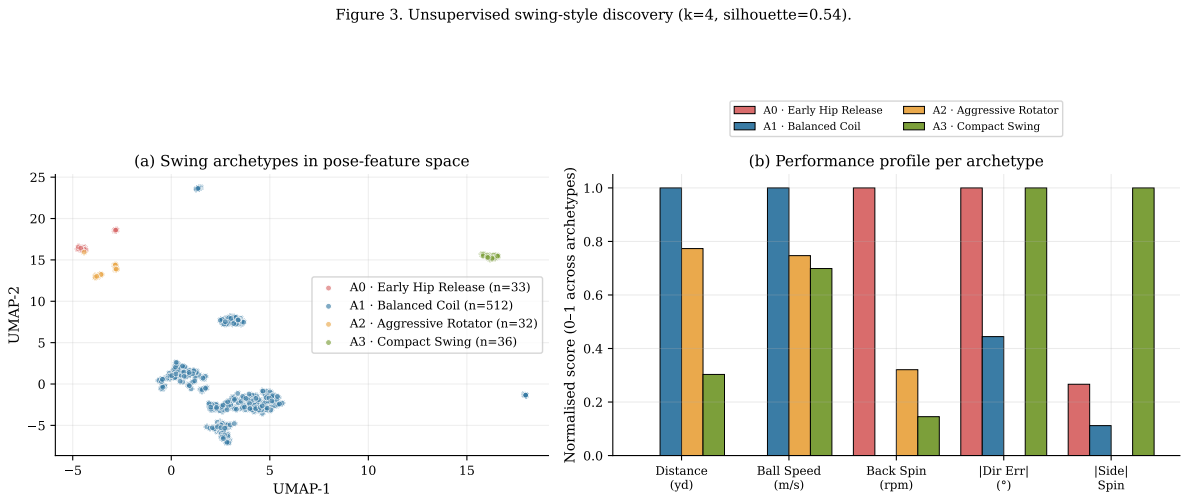

Figure 4. (a) UMAP embedding coloured by $K = 4$ K-means labels. (b) Per-archetype performance profile (min-max normalised across archetypes). A0 is uniformly dominated on power; A2 dominates on accuracy despite sub-maximal distance.

5.4 A seven-leaf coaching rule tree

Figure 5 shows a depth-3 regression tree ($R^2_{CV} = 0.34$) using only three pose features: mid-backswing lead-arm angle (`2-LEFT-ARM-ANGLE`), hip shift at the takeaway (`1-HIP-SHIFTED`), and early-downswing hip shift (`4-HIP-SHIFTED`). A reader-friendly version of the rule list:

Prescription ruleset (driver, Face-On)

`IF 2-LEFT-ARM-ANGLE > 160.9` *(i.e. keep lead arm within 19° of straight at mid-BS)*
`AND 1-HIP-SHIFTED > -0.52` *(hips not stuck behind ball at takeaway)*
`AND 4-HIP-SHIFTED > -0.15` *(hips shift toward target in early downswing)*
⇒ **predicted drive ≈201 yd** (vs. 145 yd for the worst-case leaf).

The best leaf (227 yd) additionally requires that the trail-arm angle pre-impact (`5-RIGHT-ARM-ANGLE`) be $\leq 153.5°$ — i.e. the trail arm is still bent, indicating lag is preserved into the downswing.

Figure 4. Interpretable coaching-rule tree for driver distance (depth=3, CV R^2=0.34).
Leaf values = predicted distance (yards); darker shading = longer drives.

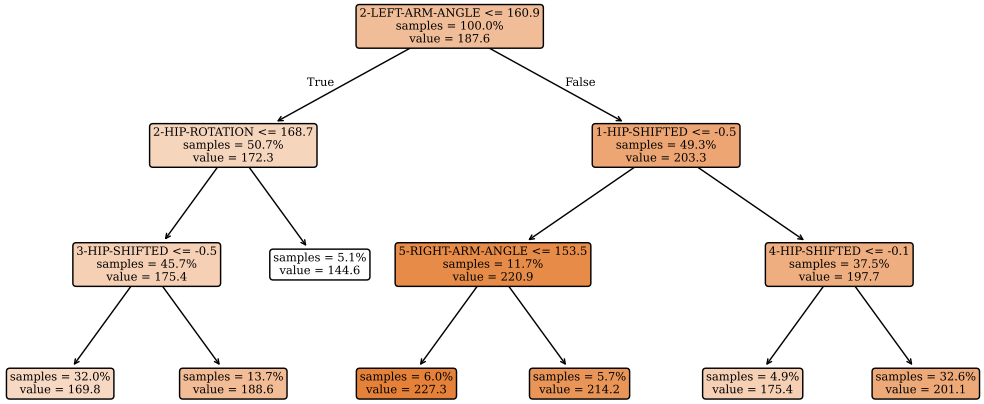

Figure 5. Depth-3 regression tree surrogate for driver distance (Face-On view). Each leaf's label is the predicted distance in yards; proportions report the share of training examples that reach each leaf.

These rules match textbook coaching language while being both data-driven and reproducible. The distribution plots in Figure 6 show that for each rule feature, the top-20% and bottom-20% distance groups are cleanly separated in modes, not just means.

6 Discussion

Why R^2 for accuracy is so much lower. The $\sim 2\times$ gap between power and accuracy ceilings (Fig. 3) is a substantive finding, not just a numerical one. Accuracy on a driver is essentially determined by club-face angle and swing path at impact — variables that the current CaddieSet features only indirectly capture. Adding clubhead-plane or face-angle signals (e.g. via a secondary high-speed camera at impact) would likely close much of the gap; without them, pose alone should be used for *distance* coaching and only as a weak accuracy signal.

From effect sizes to coaching language. Across both views, three phase-aware themes dominate the biomechanics of long drives: (i) lead-arm extension through mid-backswing, (ii) hip lateral shift toward the target in the early downswing, and (iii) preserved trail-leg flex through the top of the backswing. The depth-3 tree in Section 5.4 operationalises themes (i) and (ii) directly. Theme (iii) appears as the dominant DTL signal in Fig. 2 at phase 3 ($d = -1.40$).

Archetypes as coaching groups. The four archetypes in Section 5.3 are not just clustering artefacts: A0 and A3 appear to describe distinct faults (early hip rotation and a collapsed arm arc) while A2 is notable as an *efficient* style — not the longest, but consistently on-line. A practical use is matching a new golfer's swing to the nearest archetype and using archetype-specific rules.

Comparison with the original CaddieSet paper. Jung et al. [8] benchmark 15 expert-defined metrics and establish that interpretable predictors are consistent with domain knowledge. Our work is complementary and does not reproduce their pipeline: we use all 68 raw per-phase features directly; we rank features by effect size rather than feature importance; and we produce a decision tree whose leaves are coaching prescriptions rather than trajectory predictions.

Figure 6. Distributional shifts for the six most-discriminating pose features (long vs. short drives).

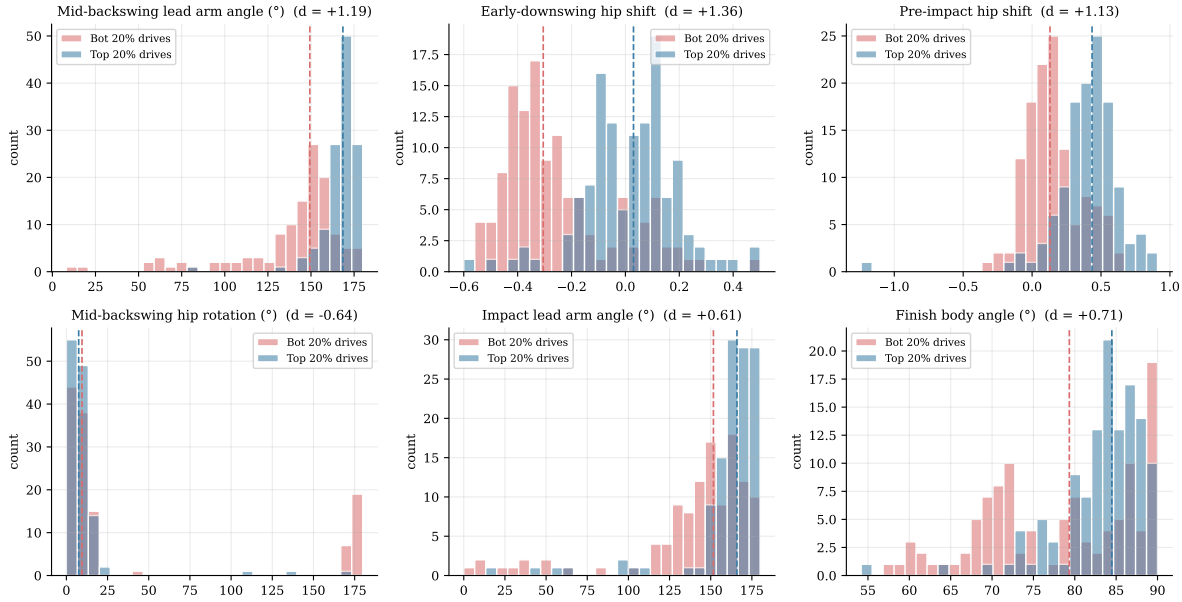

Figure 6. Distributional shifts between long (top 20%) and short (bottom 20%) drives across the six most-discriminating pose features. Dashed lines mark group medians; d = Cohen's d.

7 Limitations and Threats to Validity

- **Small golfer pool.** $N_{\text{golfer}} = 8$. Archetypes are partly golfer-specific (A3 is 89% Golfer 4; A0 is 82% Golfer 8). Replication on a larger corpus is essential before drawing population-level claims.
- **Observational data.** All associations are correlational. A straighter lead arm *could* be a cause of distance, or merely a marker of a golfer who already has the coil to support it. Interventional studies are needed to establish causal coaching value.
- **Feature-name inference.** The public dataset column names include directional descriptors (e.g. `HIP-SHIFTED`, `HANGING-BACK`) but not formal units; sign interpretation follows the most natural biomechanical reading.
- **No club-face or path data.** As noted above, the accuracy ceiling is capped by features the dataset does not include.
- **Single-view models in practice.** Real deployments will typically record only Face-On or DTL. Our cross-view complementarity result implies that either view alone leaves meaningful signal on the table.

8 Reproducibility

The code, figure-generation scripts and this manuscript source are released under an MIT licence at the GitHub repository cited on the title page. The underlying data remain the property of the authors of Jung et al. [8] and are used here under their terms. Seed values, hyperparameters, and cross-validation folds are fixed throughout; rerunning `src/05_make_figures.py` from a clean environment reproduces every figure in this paper exactly.

9 Conclusion

The CaddieSet corpus [8] is a rich substrate for biomechanical analysis well beyond trajectory prediction. By (i) shifting the question from "predict the ball" to "prescribe the swing", (ii) reporting per-phase effect sizes, and (iii) compressing the findings into a human-readable rule tree and a four-archetype clustering, we show that a small number of pose features drive most of the distance gap between long and short drives — and that the same features are *not* sufficient for predicting accuracy, motivating future sensor integration. We hope this secondary analysis encourages further prescriptive, interpretable use of the CaddieSet data.

Acknowledgements

We thank the authors of [8] for releasing the CaddieSet corpus, and the developers of `scikit-learn`, `umap-learn`, and `matplotlib` for the open-source tooling that makes this kind of secondary analysis practical.

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A Top feature tables

Table 2. Top-12 pose features ranked by |Cohen's d| between the top-20% and bottom-20% driver-distance groups, Face-On view ($n_{\text{top}} = n_{\text{bot}} = 123$; all $p < 10^{-5}$).

Feature	Top mean	Bot mean	Diff	d
4-HIP-SHIFTED	+0.018	−0.245	+0.263	+1.36
2-LEFT-ARM-ANGLE	167.9°	139.2°	+28.7°	+1.19
5-HIP-SHIFTED	+0.407	+0.155	+0.252	+1.13
7-FINISH-ANGLE	83.3°	78.0°	+5.3°	+0.71
6-RIGHT-LEG-ANGLE	159.0°	164.4°	−5.4°	−0.67
1-LEFT-ARM-ANGLE	169.1°	158.5°	+10.6°	+0.64
2-HIP-ROTATION	10.6°	43.3°	−32.7°	−0.64
3-HIP-SHIFTED	−0.447	−0.599	+0.152	+0.64
2-SHOULDER-LOC	58.1	34.9	+23.2	+0.63
3-RIGHT-LEG-ANGLE	167.3°	173.7°	−6.4°	−0.61
6-LEFT-ARM-ANGLE	161.1°	140.8°	+20.4°	+0.61
6-HIP-SHIFTED	+0.459	+0.254	+0.205	+0.54

Table 3. Top-12 pose features, Down-the-Line view ($n_{\text{top}} = 108$, $n_{\text{bot}} = 105$).

Feature	**Top mean**	**Bot mean**	**Diff**	**d**
3-RIGHT-DISTANCE	0.670	0.582	+0.088	+1.54
3-RIGHT-LEG-ANGLE	167.2°	177.6°	−10.4°	−1.40
0-SPINE-ANGLE	73.2°	77.9°	−4.7°	−1.11
3-RIGHT-ARM-ANGLE	75.8°	64.1°	+11.7°	+0.90
1-SPINE-ANGLE	75.8°	78.4°	−2.6°	−0.81
4-HIP-ANGLE	6.5°	11.2°	−4.7°	−0.76
5-LEFT-LEG-ANGLE	175.1°	169.6°	+5.6°	+0.74
6-HIP-ANGLE	12.5°	18.0°	−5.5°	−0.73
7-SHOULDER-ANGLE	7.3°	12.9°	−5.6°	−0.69
3-HIP-ANGLE	6.9°	9.7°	−2.8°	−0.63
6-SPINE-ANGLE	80.6°	77.0°	+3.7°	+0.61
2-SHOULDER-ANGLE	18.9°	21.1°	−2.2°	−0.40